

5 Electrical Characteristics

5.1 Absolute Maximum Ratings

Stresses above those listed in Table 5-1 *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only and normal operation of the device at these or any other conditions beyond those indicated in Section 5.2 *Recommended Power Supply Characteristics* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

Table 5-1. Absolute Maximum Ratings

Parameter	Description	Min	Max	Unit
VDDA, VDD3P3, VDD3P3_RTC, VDD3P3_CPU, VDD_SDIO	Allowed input voltage	−0.3	3.6	V
I_{output}^1	Cumulative IO output current	—	1200	mA
T_{STORE}	Storage temperature	−40	150	°C

¹ The product proved to be fully functional after all its IO pins were pulled high while being connected to ground for 24 consecutive hours at ambient temperature of 25 °C.

5.2 Recommended Power Supply Characteristics

Table 5-2. Recommended Power Supply Characteristics

Parameter	Description	Min	Typ	Max	Unit
VDDA, VDD3P3_RTC, VDD3P3, VDD_SDIO (3.3 V mode) <i>note 1</i>	Voltage applied to power supply pins per power domain	2.3/3.0 <i>note 2</i>	3.3	3.6	V
VDD3P3_CPU	Voltage applied to power supply pin	1.8	3.3	3.6	V
I_{VDD}	Current delivered by external power supply	0.5	—	—	A
T <i>note 3</i>	Operating temperature	−40	—	125	°C

- VDD_SDIO works as the power supply for the related IO, and also for an external device. Please refer to the Appendix [IO_MUX](#) of this datasheet for more details.
 - VDD_SDIO can be sourced internally by the ESP32 from the VDD3P3_RTC power domain:
 - When VDD_SDIO operates at 3.3 V, it is driven directly by VDD3P3_RTC through a 6 Ω resistor, therefore, there will be some voltage drop from VDD3P3_RTC.
 - When VDD_SDIO operates at 1.8 V, it can be generated from ESP32's internal LDO. The maximum current this LDO can offer is 40 mA, and the output voltage range is 1.65 V ~ 2.0 V.
 - VDD_SDIO can also be driven by an external power supply.
 - Please refer to Section 2.5.2 [Power Scheme](#), for more information.
- Chips with a 3.3 V flash or PSRAM in-package: this minimum voltage is 3.0 V;
 - Chips with no flash or PSRAM in-package: this minimum voltage is 2.3 V;
 - For more information, see Section 1 [ESP32 Series Comparison](#).
- The operating temperature of ESP32-U4WDH and ESP32-D0WDRH2-V3 ranges from −40 °C to 85 °C, due to the in-package flash or PSRAM. For other chips that have no in-package flash or PSRAM, their operating temperature is −40 °C ~ 125 °C.